

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay.
- Best CVRPLIB holdout gap: cvrp_no_replay.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_191601_e
- started_at_local: 2026-05-13 19:16:01
- finished_at_local: 2026-05-13 19:18:58
- duration_hhmm: 00:03
- duration_seconds: 176.732
- seed_offset: 4000
- replicate_label: e
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.243109	0.01914	0.143567	0.867949	0.78	0.01253
cvrp_random_replay	random	score_only	False	0.253867	0.005376	0.143427	0.910327	0.58	-0.004228
cvrp_failure_replay	failure	score_only	False	0.245177	0.005376	0.138599	0.871877	0.78	-0.001404
cvrp_failure_replay_compression	failure	novelty_gate	True	0.253827	0.005376	0.143404	0.0	0.58	0.0

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.243109, synthetic 0.01914, combined 0.143567.

- Accepted-epoch count 2, mean accepted code novelty 0.867949, and final complexity 0.78.
- Adaptation efficiency 0.01253 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.243109 across 5 instances; family means: A=0.075916, B=0.254196, E=0.324376, P=0.306859.
- Panel synthetic_holdout mean gap 0.01914 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.055054.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality_gap": 0.200893}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.253867, synthetic 0.005376, combined 0.143427.
- Accepted-epoch count 2, mean accepted code novelty 0.910327, and final complexity 0.58.
- Adaptation efficiency -0.004228 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.253867 across 5 instances; family means: A=0.143979, B=0.248809, E=0.326296, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1063, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.134472}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.245177, synthetic 0.005376, combined 0.138599.
- Accepted-epoch count 2, mean accepted code novelty 0.871877, and final complexity 0.78.
- Adaptation efficiency -0.001404 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_cvrplib mean gap 0.245177 across 5 instances; family means: A=0.091623, B=0.248805, E=0.324376, P=0.312274.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.334061}, {"best_known_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.319104}, {"best_known_cost": 784, "cost": 950, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.211735}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.253827, synthetic 0.005376, combined 0.143404.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.253827 across 5 instances; family means: A=0.115183, B=0.271742, E=0.309021, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1072, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.144077}, {"best_known_cost": 784, "cost": 858, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.094388}]

Judge Appendix

Summary of CVRP Benchmark Transfer and Optimality Gaps

Condition	Replay Mode	Final CV RPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Code Novelty (mean)	Complexity	Notes on Replay & Selection
cvrp_no_replay	None	**0.2431 (best)**	0.01914	0.14357 (3rd best)	0.868	0.78	No replay; "score_only" selection
cvrp_random_replay	Random replay	0.2539	**0.00538 (best)**	0.14343 (4th best)	**0.910**	0.58	Random replay; "score_only"
cvrp_failure_replay	Failure replay	0.2452	0.00538	**0.1386 (best)**	0.872	0.78	Failure replay; "score_only"
cvrp_failure_replay_compression	Failure replay + compression	0.2538	0.00538	0.14340	0.0	0.58	Failure replay + compression; novelty gate

Key Observations

- **Held-out CVRPLIB Gap (Primary transfer metric)****
 - Best (lowest) gap: ****cvrp_no_replay (0.2431)****
 - Failure replay (0.2452) and failure replay with compression (0.2538) slightly worse.
 - Random replay worst (0.2539).
- **Synthetic Holdout Gap****
 - Best: ****Failure replay, random replay, and failure replay compression conditions tie (~0.0054)****
 - No replay significantly worse (0.0191).
- **Transfer Gap****
 - Best: ****cvrp_failure_replay (0.1386)****
 - No replay (0.1436), random replay (0.1434), and compression-aware (0.1434) conditions are close but worse than failure replay.

4. ****Code Novelty****

- Declines to zero with compression-aware replay (0.0).
- Highest with random replay (0.91), slightly lower with no replay and failure replay (~0.87).
- Despite zero code novelty, compression-aware replay does not improve transfer metrics.

5. ****Complexity****

- Higher complexity (0.78) under no replay and failure replay.
- Lower (0.58) for random replay and failure replay + compression.

6. ****Adaptation Efficiency and Accepted Epochs****

- All conditions have low accepted epoch counts (mostly 2, except 1 with compression).
- Adaptation efficiency negative or close to zero except positive for no replay.

Conservative Interpretation

- ****No replay achieves the best generalization performance on the held-out CVRPLIB dataset, indicating stronger transfer optimality despite its higher synthetic gap.****
- ****Failure replay condition shows the best transfer gap metric, suggesting failure replay slightly improves transferring to related problem settings.****
- ****Random replay achieves the lowest synthetic gap and highest code novelty but does not lead to better held-out CVRPLIB or transfer gap performance, implying lexical/code novelty alone does not correspond to algorithmic improvement or transfer.****
- ****Compression-aware failure replay reduces code novelty drastically without improving transfer or optimality gaps, indicating that replay compression may limit beneficial adaptation.****
- ****Overall, replay modes improve synthetic holdout gap compared to no replay, but no replay still dominates held-out CVRPLIB optimality. Failure replay may offer a modest transfer advantage, albeit with similar held-out performance to no replay.****

Additional Notes

- Differences in transfer gaps among conditions are relatively small (0.1386 to 0.1436), suggesting subtle effects.
- Selection mode differs for compression-aware replay ("novelty_gate") vs others ("score_only"), which might affect code novelty and adaptation behavior.
- No evidence from metrics to support claims of algorithmic invention purely based on code novelty.

Summary Table

Condition	Held-out CVRPLIB Gap (↓ better)	Synthetic Holdout Gap (↓ better)	Transfer Gap (↓ better)	Code Novelty	Replay Type	Comments
cvrp_no_replay	**0.2431 (best)**	0.0191	0.1436	0.868	None	Best held-out performance
cvrp_failure_replay	0.2452	0.0054	**0.1386 (best)**	0.872	Failure Replay	Best transfer gap highlighting mild benefit
cvrp_random_replay	0.2539	**0.0054 (best)**	0.1434	**0.910**	Random Replay	Highest novelty but worse held-out CVRPLIB
cvrp_failure_replay_compression	0.2538	0.0054	0.1434	0.0	Failure Replay + Comp.	Compression reduces novelty, no performance gain

****Conclusion:****

- The benchmark supports the conclusion that ****no replay ensures superior held-out CVRPLIB optimality****, while ****failure replay offers a small transfer gap improvement**** without increasing novelty.
- Code novelty improvements with random replay do not translate into unequivocal transfer or optimality gains.
- Compression-aware replay sacrifices code novelty with no clear transfer benefit.
- Interpretations should focus on numeric gaps rather than code novelty or qualitative narratives of algorithmic invention.